LLM-explanations evaluation curator task instructions

https://doccano.bcgsc.ca/projects/96/guideline

The goal of this task is to evaluate how useful/correct/trustworthy the explanations generated by LLMs are.

Like previous tasks, this task will use the annotation software doccano. The current task is composed of two different subtasks which should be done simultaneously. Task #1 is generating more gold-standard reference labels and explanations. This is the same as the previous curation task. The new subtask is evaluating the responses generated by a set of LLMs.

SubTask #1: Generating Reference Labels and Explanations

For each evidence/claim pair and set of explanations you will also be asked to generate a novel label and explanation that will be used as a reference explanation for training and evaluating later models as it will be included in subsequent versions of the dataset. You can ignore the LLM responses at the top when doing this subtask. They are only needed for subtask #2. Scroll down to the claim/evidence at the bottom to start reviewing the evidence. Please use formal language and avoid uncommon abbreviations or slang. This label and explanation should be added using the “comment on document” function and follow the format “<LABEL>. <EXPLANATION>”.

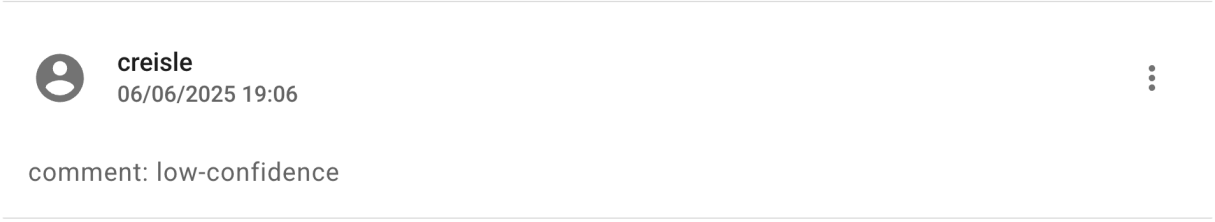
See the previous task instructions for examples on what appropriate labels and explanations could look like. Note that in contrast to the previous task, the label and explanation will be entered in the “**comment**” field this time since the main entry area will be used for subtask #2.

Note: If you think an example is close to SUPPORTS or REFUTES but is missing some information you can label it NEI/SUPPORTS or NEI/REFUTES. The gold label will still be NEI but this will make it obvious that there is partial support or partial refutes.

ex. “NEI/REFUTES. The evidence describes sensitivity in patients with low BRCA1 expression when treated with platinum compounds which contradicts the claim of resistance. However the disease group in the evidence is sporadic epithelial ovarian cancer which is more specific than the disease specified by the claim, ovarian cancer.”

any comments you want to add that are not the answer to subtask #1, add as a separate comment, and just preface them with “comment:” so they are easy to remove when I am looking for the stance + explanation. For example, If you are particularly

unsure of evaluating a particular claim/evidence pair (ex. you feel you have insufficient understanding or expertise of the disease or experiments done) you can add another comment with a low-confidence label like so



SubTask #2: Evaluating the Quality of LLM-generated Responses

The models were given a claim/evidence pair and asked to generate a label and explanation. You will be asked to tag/highlight the sentences (or parts of sentences, or spans of sentences) for each explanation with the following categories

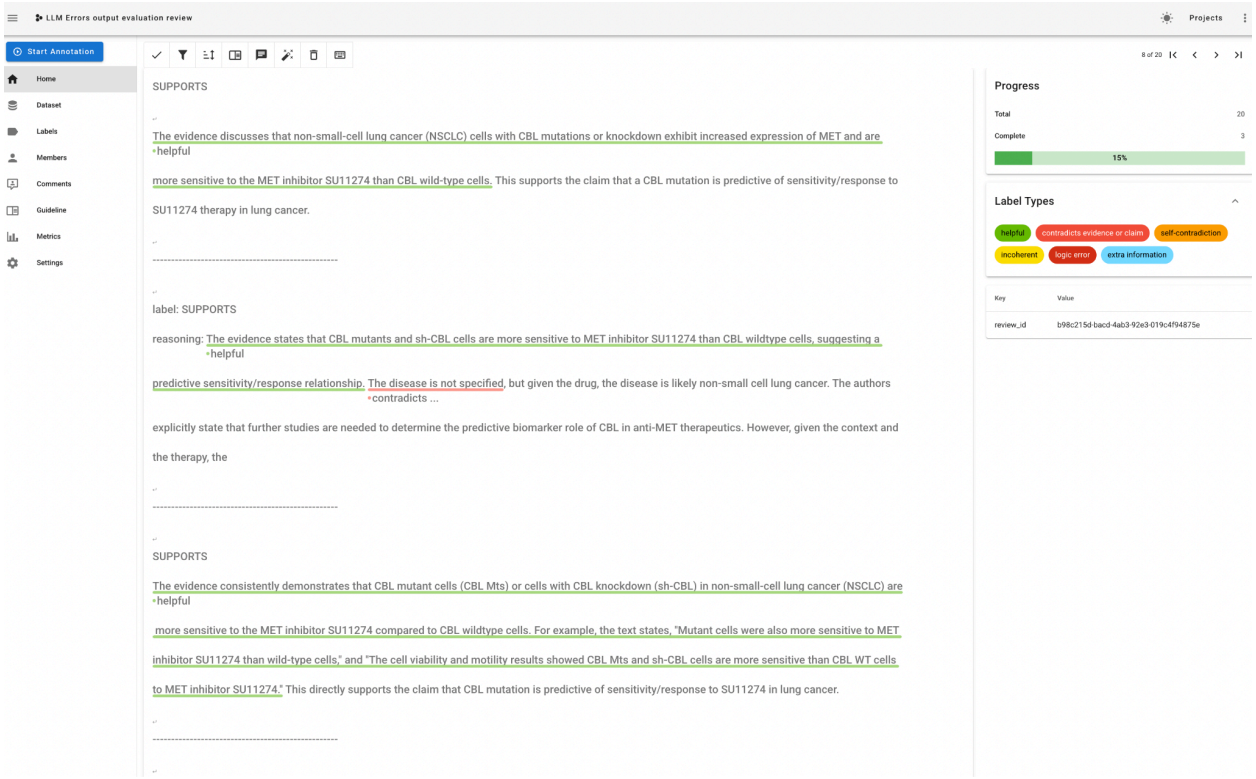
- helpful
    - The selected text is directly relevant to the claim/evidence pair in question and is useful and specific in explaining why/how the stance of the evidence is supports, refutes, or not enough information toward the claim. Should go beyond simply restating the claim.
- contradicts evidence or claim (external consistency)
    - The selected text is not consistent with the facts of the evidence or claim
- self-contradiction (internal consistency)
    - Within the explanation (or between the generated label and explanation), the model contradicts itself
- incoherent
    - The selected text is not understandable or very unclear and therefore cannot be evaluated. This could mean a paragraph that is worded in such a way as to be very difficult to understand, not necessarily impossible.
- logic error
    - The selected text makes a logical leap, assumption, or inference that is obviously incorrect
- extra information
    - The selected text is not from the evidence or claim that was provided. This could include information or citations from other papers or sources, parts of the paper that the evidence is from but that were not included in the evidence provided to the model, or even hallucinations

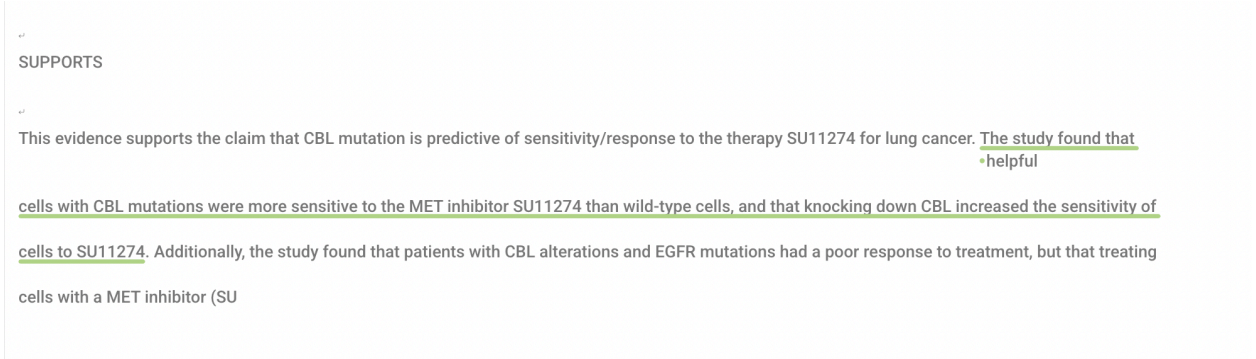
To avoid curators having to re-read the evidence/claim pairs for evaluating multiple responses to the same evidence/claim pairs the responses will be grouped, delimited by a line of dashes

within the text field (see figure below). There will be responses from 5 different LLMs and the order is randomized each time.

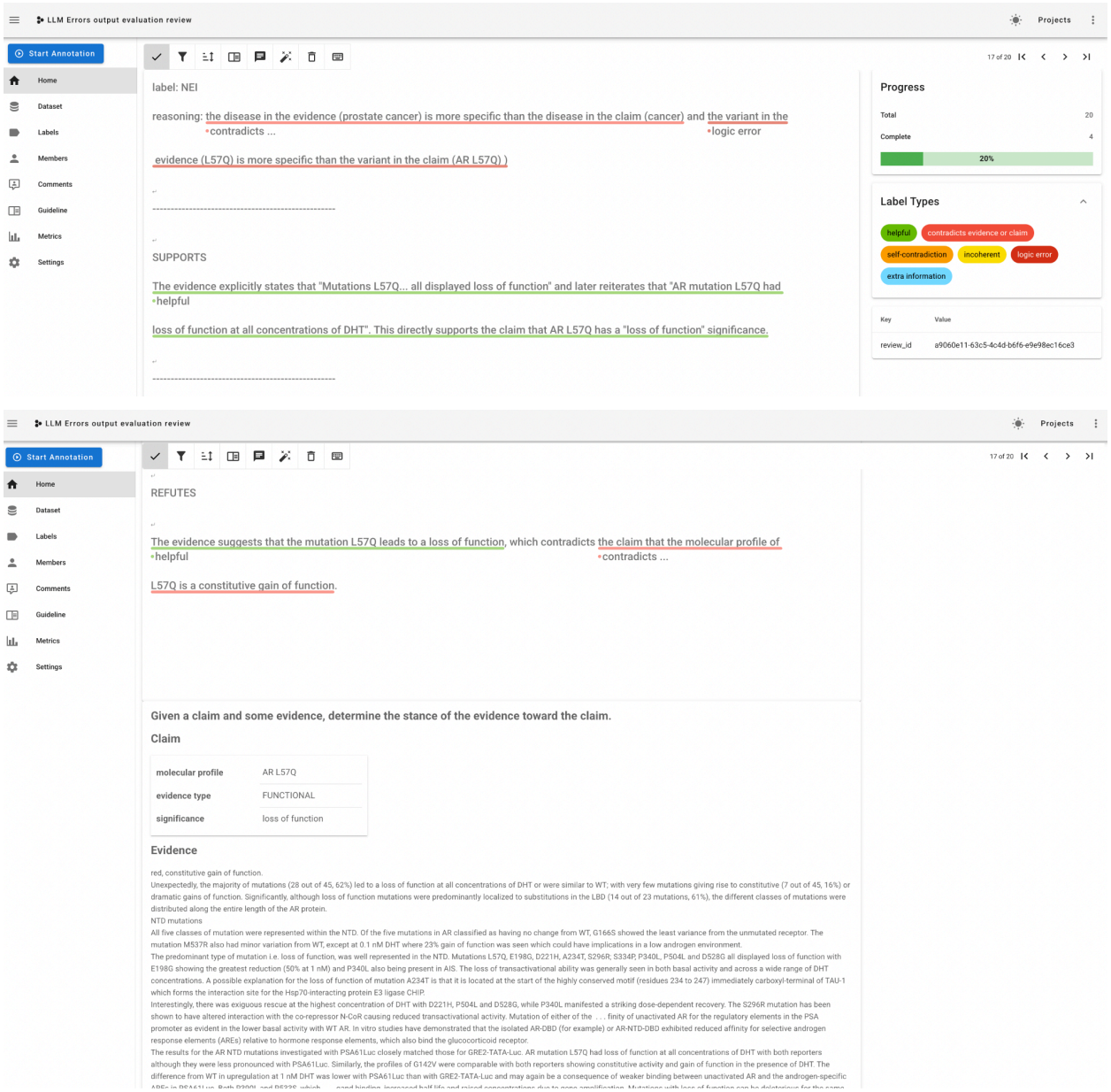
Example #1: Disease in evidence is more specific than disease in claim.

We select/tag the parts of the evidence that are helpful in green but we do not select/tag irrelevant information. In red, we select/tag the part of the response which states the disease was not specified because the evidence did in fact specify the disease.





Example #2


LLM Errors output evaluation review
Projects
Start Annotation
Home
Dataset
Labels
Members
Comments
Guideline
Metrics
Settings
17 of 20
label: NEI
reasoning: the disease in the evidence (prostate cancer) is more specific than the disease in the claim (cancer) and the variant in the evidence (L57Q) is more specific than the variant in the claim (AR L57Q))
•contradicts ...
•logic error
SUPPORTS
The evidence explicitly states that "Mutations L57Q... all displayed loss of function" and later reiterates that "AR mutation L57Q had loss of function at all concentrations of DHT". This directly supports the claim that AR L57Q has a "loss of function" significance.
•helpful
Progress
Total 20
Complete 4
20%
Label Types
helpful
contradicts evidence or claim
self-contradiction
incoherent
logic error
extra information
Key
Value
review_id
a9060e11-63c5-4c4d-b6f6-e9e98ec16ce3
LLM Errors output evaluation review
Projects
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REFUTES
The evidence suggests that the mutation L57Q leads to a loss of function, which contradicts the claim that the molecular profile of L57Q is a constitutive gain of function.
•helpful
•contradicts ...
Given a claim and some evidence, determine the stance of the evidence toward the claim.
Claim
molecular profile
AR L57Q
evidence type
FUNCTIONAL
significance
loss of function
Evidence
red, constitutive gain of function.
Unexpectedly, the majority of mutations (28 out of 45, 62%) led to a loss of function at all concentrations of DHT or were similar to WT; with very few mutations giving rise to constitutive (7 out of 45, 16%) or dramatic gains of function. Significantly, although loss of function mutations were predominantly localized to substitutions in the LBD (14 out of 23 mutations, 61%), the different classes of mutations were distributed along the entire length of the AR protein.
NTD mutations
All five classes of mutation were represented within the NTD. Of the five mutations in AR classified as having no change from WT, G166S showed the least variance from the unmutated receptor. The mutation M537R also had minor variation from WT, except at 0.1 nM DHT where 23% gain of function was seen which could have implications in a low androgen environment.
The predominant type of mutation i.e. loss of function, was well represented in the NTD. Mutations L57Q, E198G, D221H, A234T, S296R; S334P, P340L, P504L and D528G all displayed loss of function with E198G showing the greatest reduction (50% at 1 nM) and P340L also being present in AIS. The loss of transactivational ability was generally seen in both basal activity and across a wide range of DHT concentrations. A possible explanation for the loss of function of mutation A234T is that it is located at the start of the highly conserved motif (residues 234 to 247) immediately carboxyl-terminal of TAU-1 which forms the interaction site for the Hsp70-interacting protein E3 ligase CHIP.
Interestingly, there was exiguous rescue at the highest concentration of DHT with D221H, P504L and D528G, while P340L manifested a striking dose-dependent recovery. The S296R mutation has been shown to have altered interaction with the co-repressor N-CoR causing reduced transactivational activity. Mutation of either of the ... finity of unactivated AR for the regulatory elements in the PSA promoter as evident in the lower basal activity with WT AR. In vitro studies have demonstrated that the isolated AR-DBD (for example) or AR-NTD-DBD exhibited reduced affinity for selective androgen response elements (AREs) relative to hormone response elements, which also bind the glucocorticoid receptor.
The results for the AR NTD mutations investigated with PSA61Luc closely matched those for GRE2-TATA-Luc. AR mutation L57Q had loss of function at all concentrations of DHT with both reporters although they were less pronounced with PSA61Luc. Similarly, the profiles of G142V were comparable with both reporters showing constitutive activity and gain of function in the presence of DHT. The difference from WT in upregulation at 1 nM DHT was lower with PSA61Luc than with GRE2-TATA-Luc and may again be a consequence of weaker binding between unactivated AR and the androgen-specific

Other Examples

label: SUPPORTS. This seems to involve NOTCH gain of function mutations.

Evidence is very supporting.

label: REFUTES. As shown in the cited study, NOTCH1 was found not to be extra inform...

mutated in this leukemia: "The leukaemia cells of 48 cases, including ALL,

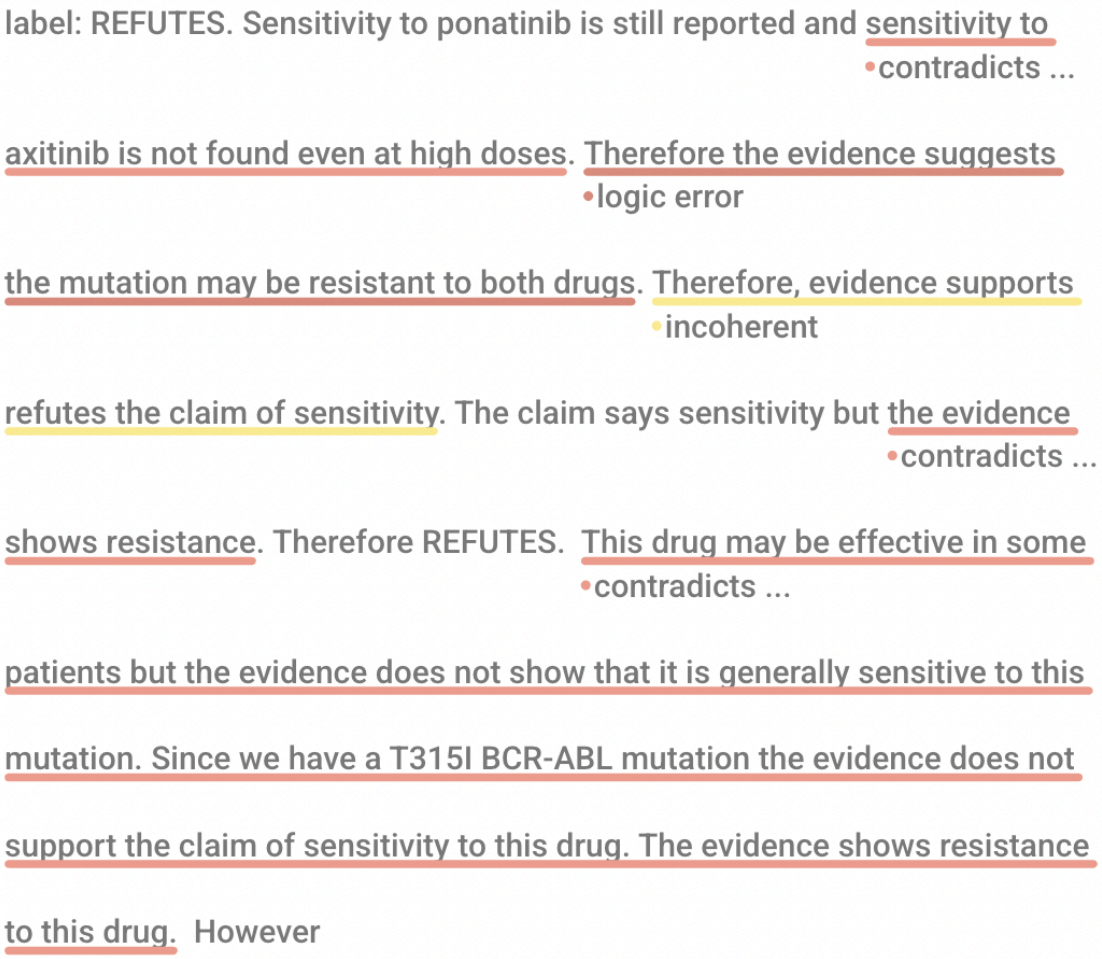
were tested by three different NOTCH1 fusion detection assays, but none

were positive." Hence this seems to be a negative example of leukemia

associated with NOTCH mutation.

label: NEI. In the context of T-ALL, there have been many gene fusions

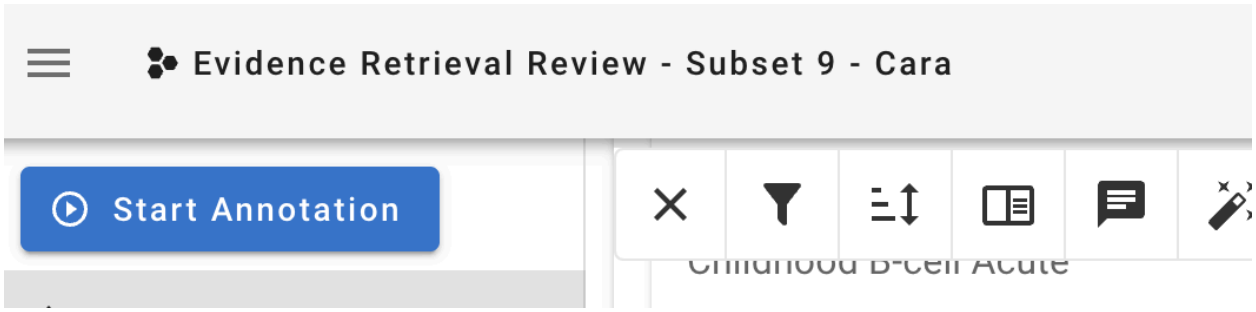
leading to Notch1 overexpression,



Notes on Using Doccano

Marking your Progress

To ensure your count goes up as you progress through the entries, the checkbox at the top left must be changed from X to a checkmark by clicking it. This is useful if you wish to only check the ones you are sure about and leave it unchecked for ones you want to come back to later



FAQ

Q. Are the responses always in the same order?

A. no the order of responses is randomized for each example as to not bias the annotation by position

Q. Do we have to highlight all the text?

A. No, just select text that corresponds to one of the tag types discussed. Some text can be left blank if it is irrelevant, or otherwise does not fit into the tags

Q. Do we need to comment about why we applied a tag to a span of text?

A. No, the comment section should exclusively be used for generating the gold standard label + explanation

Q. A number of the LLM statements seem to be cut off mid-sentence. I started tagging those as 'incoherent' but maybe I should just leave them blank if they're at the end?

A. Yes, leave cut-off sentences at the end blank. if it is a multi-part sentence you can still tag the sentence fragment that is complete.

Q. If the label that the LLM gives is wrong (supports when it should be NEI), and it makes a statement about that label (eg. "the evidence supports the claim that ..."), is that a logic error because it has made an incorrect inference? Or is that contradicting the evidence or claim (in this case, it matches the claim, but doesn't completely match the evidence because the tumour type is too broad)

A. Generally I just leave statements where the LLM repeats the label + claim as blank since they are redundant when we can automatically mark the bad label as wrong.

Q. One of the LLMs is very verbose. I'm still tagging it all helpful even if it's three sentences as long as it's all directly relevant and correct, is that reasonable? Or would you rather the minimum be labelled helpful?

A. yes, any helpful sentence should be tagged helpful regardless of the length of the overall response

Q. What do I do if two sentences say the same thing in two ways and they are totally redundant?

A. You can just highlight/tag the more information-rich one and leave the other blank

Q. What if the statement says things that are not "contradictory", but are ultimately wrong. Say it is getting 99 percent of the evidence right but sort of hallucinates one small but important detail. And that detail that it is making up could be correct and consistent, not contradictory, to the given evidence. How do you want this scenario tagged?

A. any hallucinations should be tagged "extra information"

Q. How to tag a correct statement in the context of an incorrect evaluation. Say the LLM has incorrectly said SUPPORTS when it is not. But there are some things that the LLM correctly describes from the evidence, that are consistent with "Supports". Like concrete hypothetical is as follows: Say the Evidence says patients with disease A and Variant X are sensitive to drug alpha. And then the Claim says patients with disease A and Variant X are sensitive to drug beta. This would be NEI. Then in one of the LLM texts, the LLM also calls it as Supports, and in the text it says this sentence: "The evidence shows us that patients with disease A and Variant X are sensitive to a drug". How would you tag this correct sentence?

A. without specifying the drug, and there being multiple drugs in the evidence, it's pretty useless to us so I would leave it blank as irrelevant

Q. If a statement is just wrong, how would you tag that? If it says for instance "cancer is a disease only observed in mice", how would you tag this statement?

A. I haven't seen any quite like that but I use logic error when things are pretty obviously wrong but not directly contrasting the evidence or claim etc

Q. what if a statement overstates something correct. Like say the evidence mentioned a thing X. If the statement from the LLM then says that the thing X appears a lot in the evidence, and that lots of evidence is given in the evidence section for this thing X. Those last two statements are not correct, but thing X still remains true and mentioned in the text. How do I tag statements that make false claims about the extent of how much thing X is mentioned and supported by evidence?

A. if the specific details of the sentence are helpful still but just the quantifier is wrong/misleading, then you could highlight/tag the sentence as helpful but highlight/tag the quantifier (ex. lots) as contradicts evidence or claim.

Q. If the evidence indicates that Mutation X is resistant to Drug D in Cancer A, and the Claim says Mutation X is sensitive to Drug D in Cancer A then we obviously choose REFUTES. if the evidence indicates that Mutation X is resistant to Drug D in Cancer A, and the Claim says Mutation X is sensitive to Drug D in Cancer B, then do we still choose REFUTES or is this NEI, since not enough information is given about how this would work in Cancer B?

A. generally I would choose NEI or NEI/REFUTES here, which indicates that the evidence partially refutes the claim. NEI/REFUTES will end up as NEI in the dataset for now (we need a lot of these before we can use more than 3 labels) but it gives us more information on your interpretation. Depending on how similar disease A and B are may change your interpretation slightly too

Q. How should we tag wrong statements that are not part of the evidence? Is that Extra? So can Extra be information from elsewhere that is correct or made up and incorrect? Since the instructions say hallucinations can be extra, then wrong statements should be tagged Extra from what I understand

A. anything not from the evidence or claim that is not general knowledge should be tagged extra information regardless of the accuracy of the statement. We will do a separate step later to investigate all things tagged extra information to determine if they come from things the LLM has memorized or are true hallucinations but that is outside the scope of the current task since it will be very time consuming

Q. If the label is refutes and the evidence sensitive but the claim says resistant. I highlight/tag AI sentence as helpful when it says that the evidence says sensitive. Right?

A. you can evaluate sentences/fragments individually. It doesn't matter if other parts of the response are contradictory or the label is wrong. If the sentence/fragment you are

tagging is helpful and specific to the claim/evidence pair that was given as input you would tag it as helpful. It is is super non-specific however, saying something like "the evidence supports the claim" without any details, I would not consider that helpful generally since it doesn't say how it supports anything, in those cases I often leave it blank

Q. For my logic, if the statement was labeled as Refutes. I'm still tagging what the sentences are relative to the original claim and not the label? Or relative to the label?

A. the "relative to" depends on what tag we are talking about (see tag descriptions in the task description). So if you are tagging, "self-contradiction" then you are tagging relative to the label in the response as well as any other sentences in the same response. If you are tagging "contradicts evidence or claim" then you should be looking for contradictions the response states relative to the claim/evidence pair that it was given as input.

Q. What if the LLM makes an inference or generalization based on the evidence that is not wrong but leads to the incorrect label? Does that count as a hallucination/extra?

A. This is hard to say without seeing the actual example, but you might tag this a logic error. Like I have tagged a set of two sentences as a logic error when it was something like "The evidence states that variant X is often seen in disease Y. Therefore the evidence supports that variant X is a loss of function"

Q. Do we need to tag all sentences/parts of the responses?

A. No, any content that does not fit with one of the tags described in subtask #2 instructions can be left blank. Often this will be content that may be accurate but is irrelevant to the current claim/evidence pair. In rare cases this could even mean an entire LLM response may be left blank